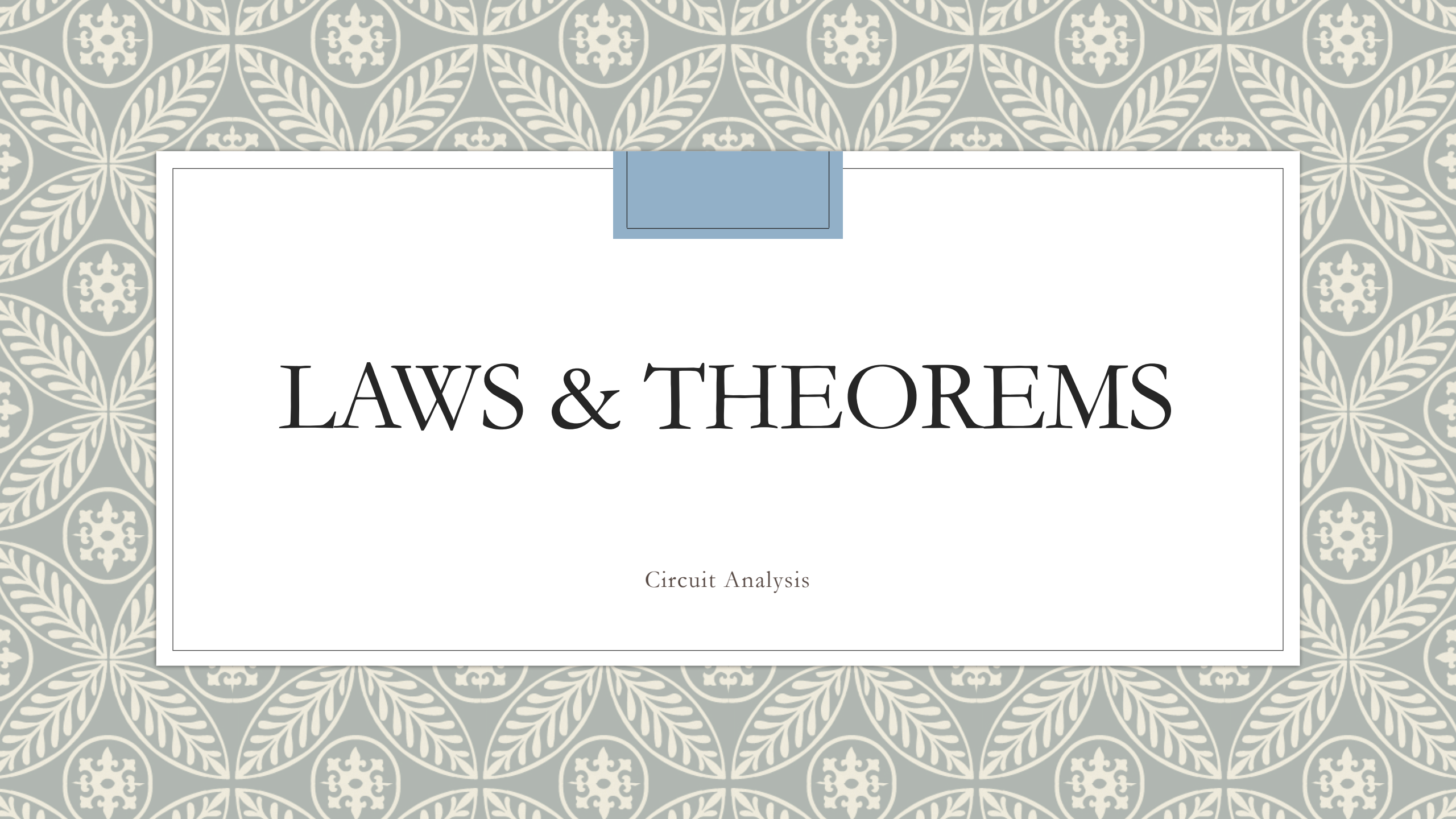
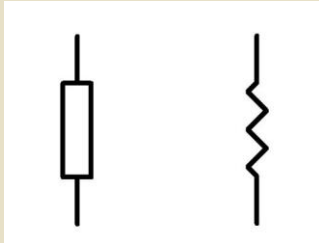


LAWS & THEOREMS

Circuit Analysis

Resistance & Resistor

- Materials have inherent characteristics to resist the flow of electric charge. This physical property to resist current is known as resistance. Generally denoted by R and measured in Ω (ohm). (Named after Georg Simon Ohm)
- Resistance of a material is $R = \rho \frac{L}{A}$; Where L = Length and A = Uniform cross-sectional area of that material.
- ρ = Resistivity of the material in ohm-meters. Good conductors have low resistivity. Insulators have high resistivity.
- The circuit element that shows resistance is called a resistor.
- Resistors are generally made of metallic alloys and carbon compounds.
 - Symbol of resistor: 
 - More Symbols and Types are available: To know more visit: <http://tinyurl.com/26xnrad4>

Resistivities of common materials.

Material	Resistivity ($\Omega \cdot m$)	Usage
Silver	1.64×10^{-8}	Conductor
Copper	1.72×10^{-8}	Conductor
Aluminum	2.8×10^{-8}	Conductor
Gold	2.45×10^{-8}	Conductor
Carbon	4×10^{-5}	Semiconductor
Germanium	47×10^{-2}	Semiconductor
Silicon	6.4×10^2	Semiconductor
Paper	10^{10}	Insulator
Mica	5×10^{11}	Insulator
Glass	10^{12}	Insulator
Teflon	3×10^{12}	Insulator

No Need to memorize this table

Conductance

- The reciprocal of resistance R is known as conductance. It is denoted by G . Conductance is the ability of an element to conduct electric current.
- Mathematically: $G = \frac{1}{R} = \frac{I}{V}$
- Unit: It is measured in $\mathfrak{U}$ (mho), or siemens (S), or Ω^{-1}
- $1 \text{ S} = 1 \mathfrak{U} = 1 \text{ A/V}$
- $I = GV$; $P = VI = V^2G = \frac{I^2}{G}$

Ohm's Law

Statement: The electric current through a conductor between two points is directly proportional to the voltage across the two points.

Mathematically: $I \propto V$; or, $I = \frac{1}{R} \times V$; or, $V = IR$

Here, V= Applied voltage between two points, I= Current through it, R= Resistance of that material.

From the equation, $1\Omega = 1 \text{ V/A}$

- In short circuit condition $R=0$, then $V = IR = 0$ (A Short circuit is a circuit element with a resistance approaching zero). Thus voltage is zero, but the current could be anything.
- In open circuit condition, $R=\infty$, then $I = \frac{1}{R} \times V = 0$ (A open circuit is a circuit element with resistance approaching infinity). Thus the current is zero, but voltage could be anything.



Limitations of Ohm's Law

Ohm's law cannot be applied to a/an-

1. Unilateral networks.
2. Non-linear objects.
3. Non-metallic conductors,
4. Complicated circuits.

However, it is not necessary that all devices follow ohm's law. There are many non-ohmic devices.

Basic problems to practice

1. An electric iron draws 2 A at 120 V. Find its resistance. (Ans: 60 ohm)
2. The essential component of a toaster is an electrical element (a resistor) that converts electrical energy to heat energy. How much current is drawn by a toaster with resistance 15 ohm at 110 V? (Ans: 7.333A)
3. A circuit consists of 30V voltage source and 5 k Ω resistor. Calculate the current i , the conductance G , and the power p . (Ans: 6 mA, 0.2 mS, 180 mW)

Above math's are only for MCQ

Branches, Nodes, and Loops

Branch: A branch represents a single element such as a voltage source or a resistor. or, any two terminal device is a branch. In the figure, there are 5 branches, namely, Voltage source, current source and three resistor.

Node: A node is the point of connection between two or more branches.

If a single wire connects two nodes, then these two nodes becomes a single node.

In the figure, there are three nodes namely a, b and c nodes.

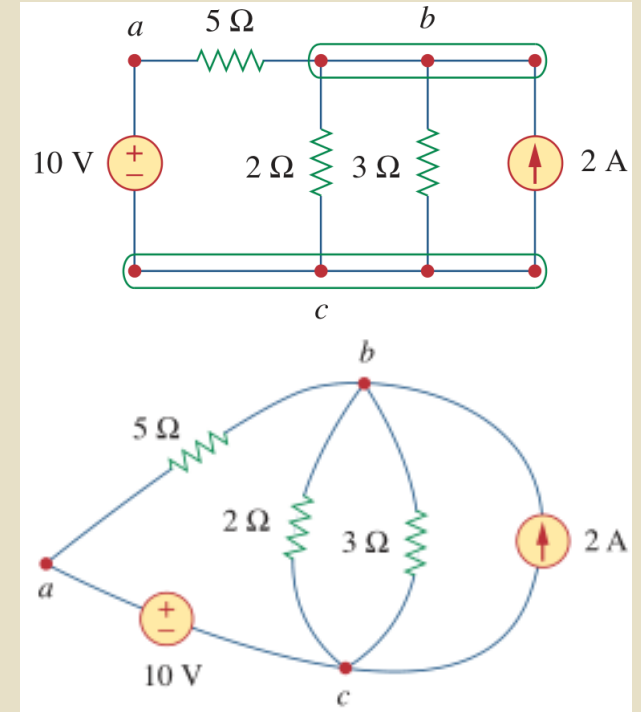
Loop: A loop is any closed path in a circuit.

A loop is a closed path formed by starting at a node, passing through a set of nodes, and returning to the starting node without passing through any node more than once.

A loop is said to be **independent** if it contains at least one branch which is not a part of any other independent loop.

Independent loops or paths result in independent sets of equations.

Here, Loop abca and loop bc through current source are independent loop. But bc from 2 ohm and 3 ohm is dependent loop



Loops Contd...

- A network with b branches, n nodes, and l independent loops will satisfy-

$$b = l + n - 1$$

Two or more elements are in series if they exclusively share a single node and consequently carry the same current.

If they are chain connected or connected sequentially, end to end.

If they share one common node and no other element is connected to that common node.

Two or more elements are in parallel if they are connected to the same two nodes and consequently have the same voltage across them.

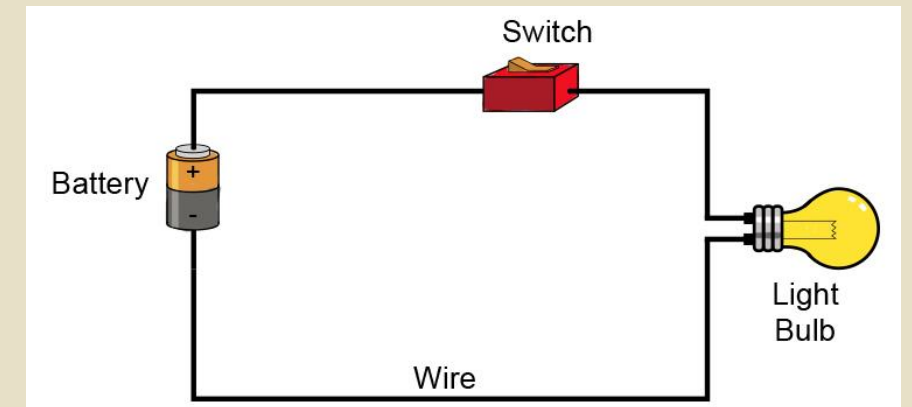
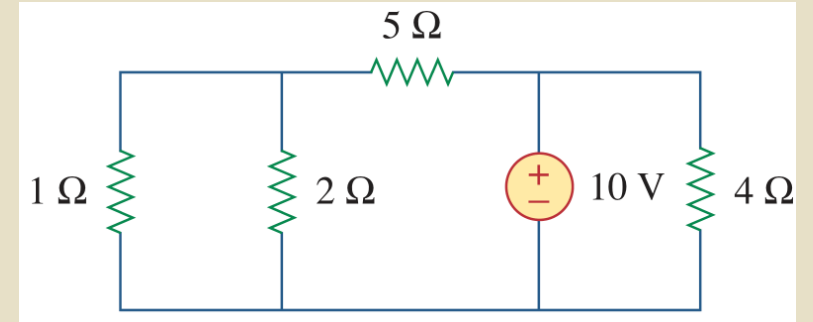
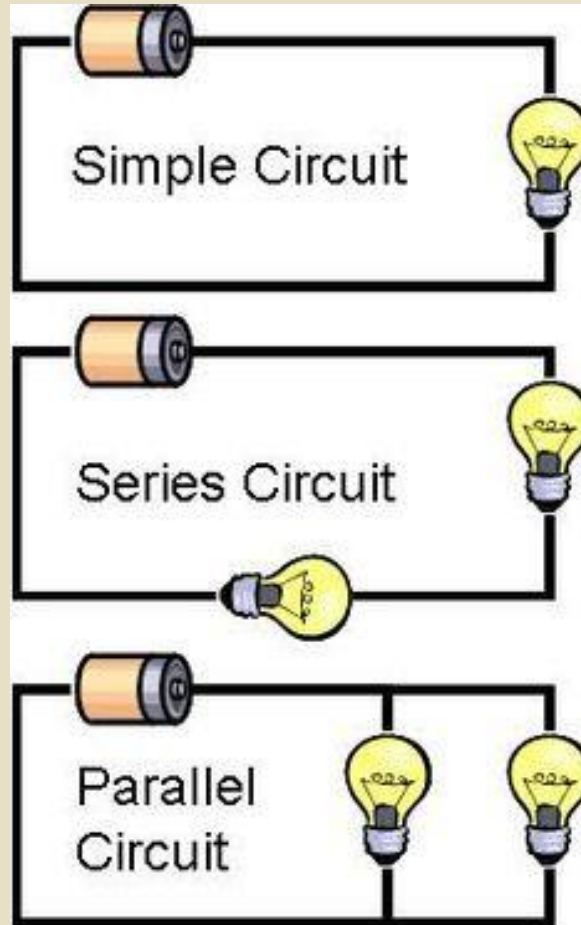
Elements can be connected in a way that they are neither in series nor in parallel. We will study about this further.

-

In the previous circuit, $5\ \Omega$ resistor is in series with 10V voltage source. And $6\ \Omega$ resistor is in parallel with 2A current source.

Practice

1. How many branches and nodes does the circuit have? Identify the elements that are in series and in parallel.



Kirchhoff's Laws

- German physicist Gustav Robert Kirchhoff has given 2 laws namely KVL and KCL based on the law of conservation of charge, which requires the algebraic sum of charges within a system cannot change.

Kirchhoff's current law (KCL): "The algebraic sum of currents entering a node (or a closed boundary) is zero.

Mathematically, $\sum_{n=1}^N i_n = 0$;

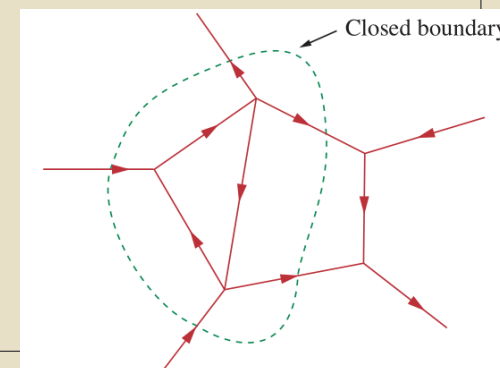
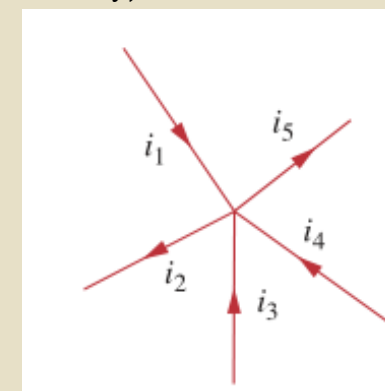
where, N = the number of branches connected to the node.

i_n = n^{th} current entering or leaving the node.

Notation: Current entering a node- regarded as positive; current leaving a node- regarded as negative.

In a closed boundary, the sum of the currents entering a node is equal to the sum of the currents leaving the node. For the circuit, $i_1 + i_3 + i_4 = i_2 + i_5$.

A node is a shrunk closed boundary. Another form of closed boundary is shown in the fig.



Kirchhoff's Voltage Law (KVL)

Statement: The algebraic sum of all voltages around a closed path (or loop) is zero.

Mathematically: $\sum_{m=1}^M v_m = 0$; where, M is the number of voltages/ branches in the loop & $v_m = m^{\text{th}}$ voltage.

Notation: Current entering a sink is +ve signed and current leaving a sink is -ve signed.

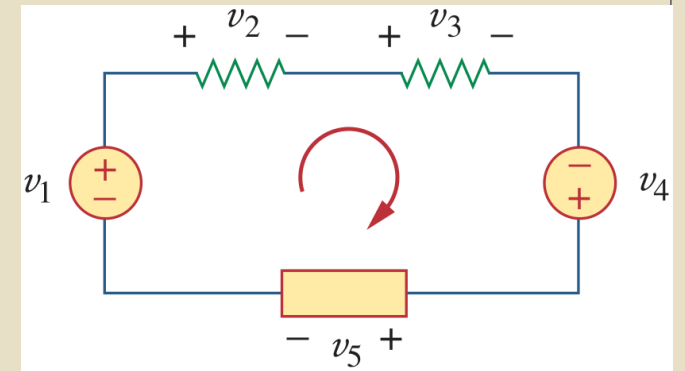
Current flowing according to source = +ve to -ve

Generally we consider clockwise current. If we consider counterclockwise, directions will be altered.

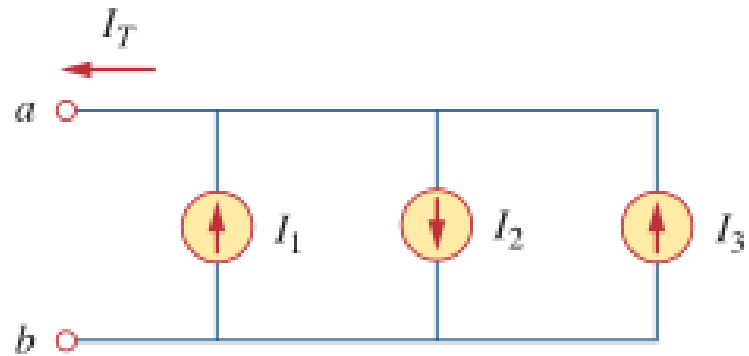
For the given circuit, $-v_1 + v_2 + v_3 - v_4 + v_5 = 0$

$$\text{or, } v_2 + v_3 + v_5 = v_1 + v_4$$

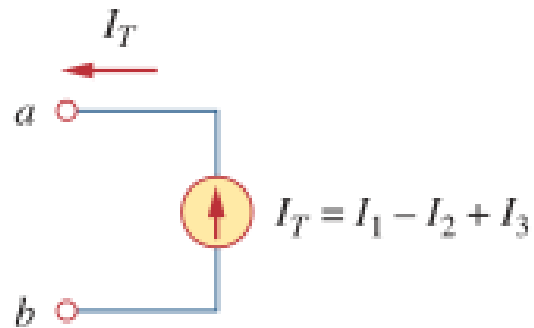
Thus, Sum of voltage drops = Sum of voltage rises



KVL and KCL

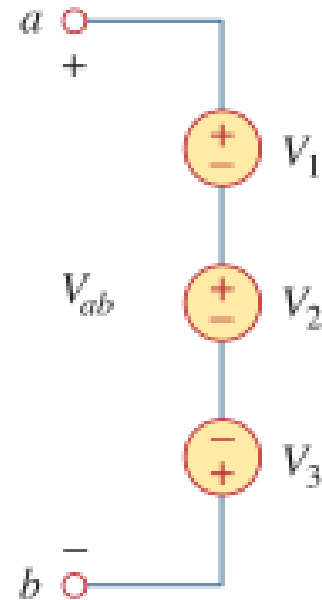


(a)

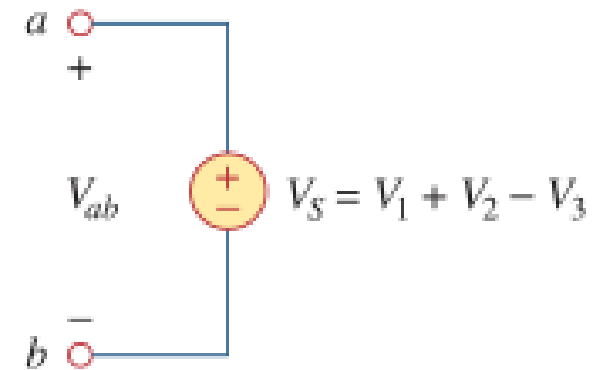


(b)

Current sources in parallel: (a) original circuit, (b) equivalent circuit.



(a)

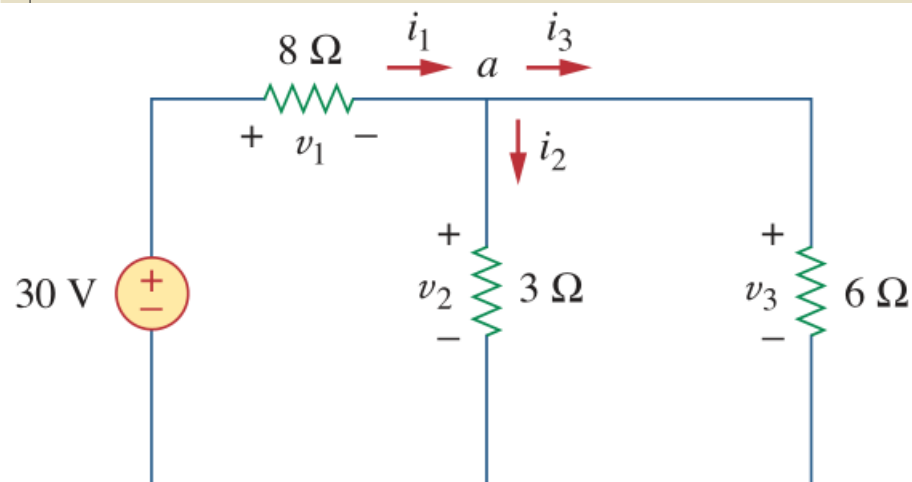
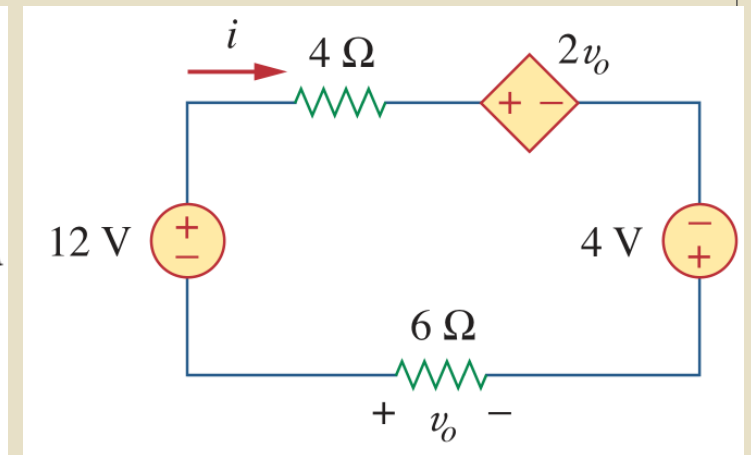
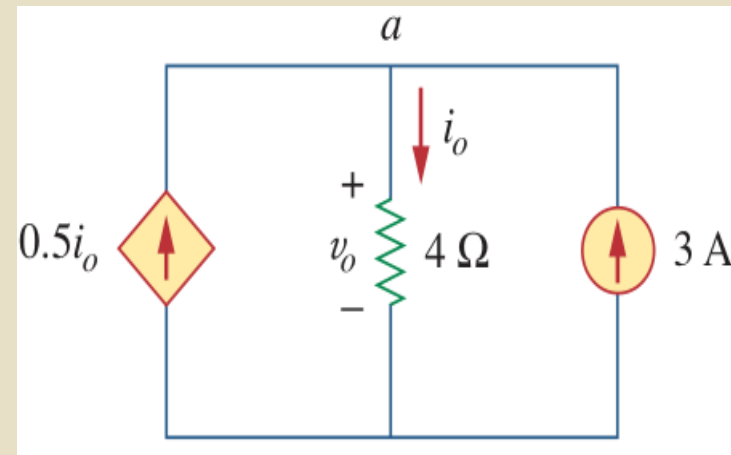
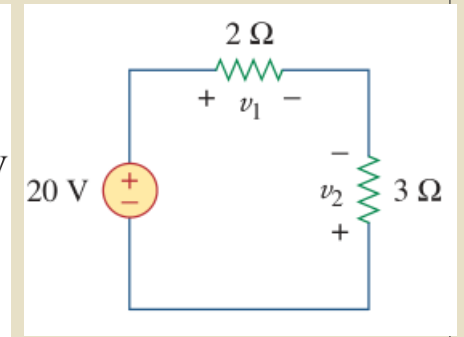
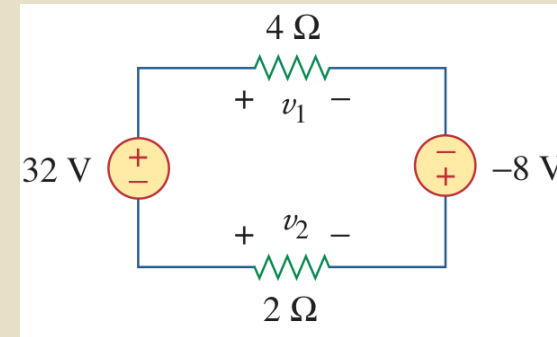


(b)

Voltage sources in series: (a) original circuit, (b) equivalent circuit.

Practice (Just for learning not for exam)

1. Find voltages v_1 and v_2 for the first row circuits.
2. Find v_0 and i for the circuit in second row.
3. Find current and voltages from the circuit shown below at left bottom corner.



Series Resistors and Voltage Division

The equivalent resistance of any number of resistors connected in series is the sum of the individual resistances.

For N resistors in series, $R_{eq} = R_1 + R_2 + \dots + R_N = \sum_{n=1}^N R_n$

For the given circuit, $v_1 = iR_1$; $v_2 = iR_2$; $-v + v_1 + v_2 = 0$

$v = v_1 + v_2 = i(R_1 + R_2)$; or, $i = \frac{v}{R_1 + R_2}$; Thus, $R_{eq} = R_1 + R_2$

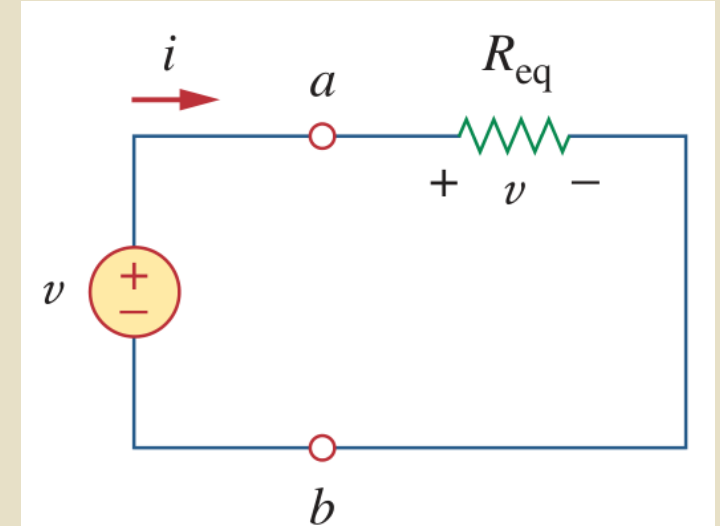
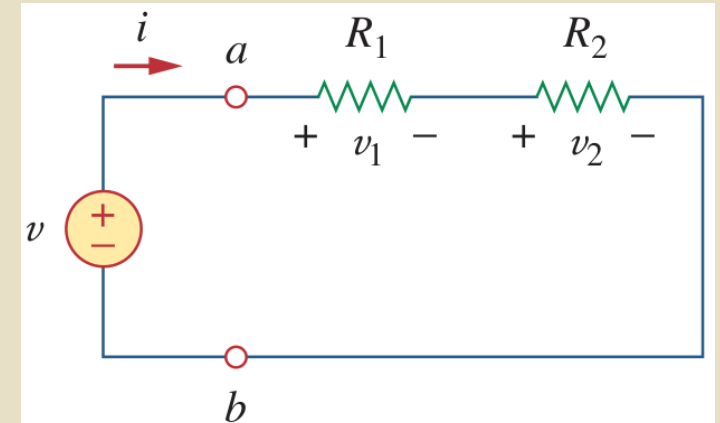
Voltage across each resistor,

$$v_1 = \frac{R_1}{R_1 + R_2} v; \quad v_2 = \frac{R_2}{R_1 + R_2} v$$

The source voltage v is divided among the resistors in direct proportion to their resistance. This is called principle of voltage division and given circuit is called voltage divider circuit.

If n resistor exists in series,

$$v_n = \frac{R_n}{R_1 + R_2 + \dots + R_N} v$$



Parallel Resistors and Current Division

The equivalent resistance of two parallel resistors is equal to the product of their resistances divided by their sum.

For the circuit, $v = iR_1 = iR_2$; $i_1 = \frac{v}{R_1}$, $i_2 = \frac{v}{R_2}$

From KCL, $i = i_1 + i_2 = \frac{v}{R_1} + \frac{v}{R_2} = v \left(\frac{1}{R_1} + \frac{1}{R_2} \right) = \frac{v}{R_{eq}}$

Thus, $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{R_1 + R_2}{R_1 R_2}$

We find that, $R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$

For N resistor in parallel,

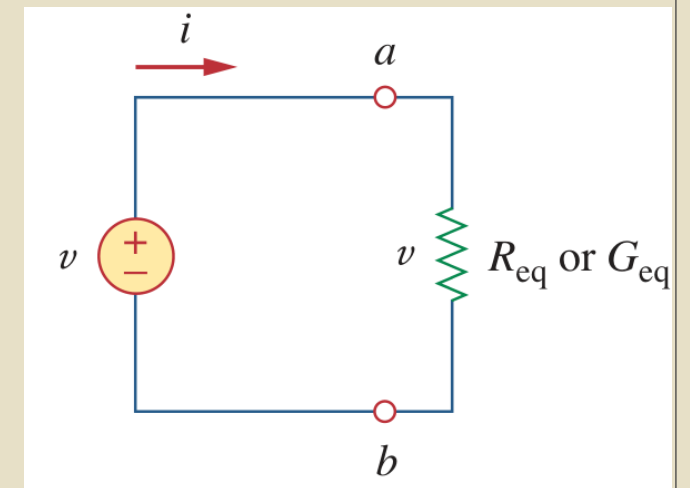
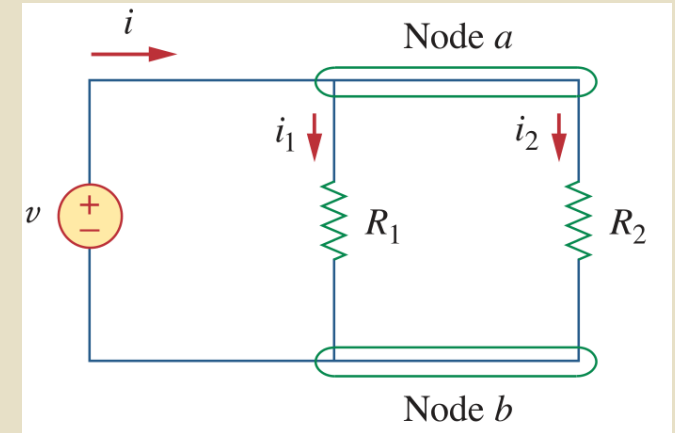
$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_N}$$

Note: R_{eq} is always smaller than the resistance of the smallest resistor in the parallel combination. If $R_1 = R_2 = \dots = R_N = R$; Then $R_{eq} = R/N$

As conductance is inverse of resistance, thus

$$G_{eq} = G_1 + G_2 + \dots + G_N$$

In parallel, equivalent conductance is the sum of individual conductance.



Current Division

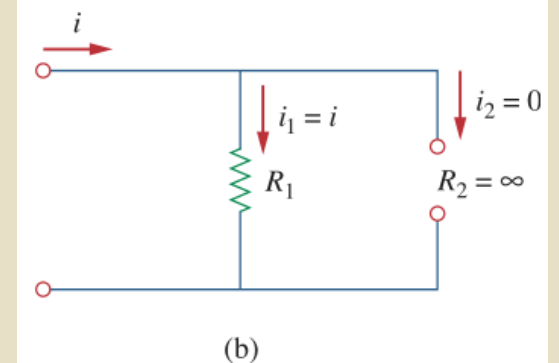
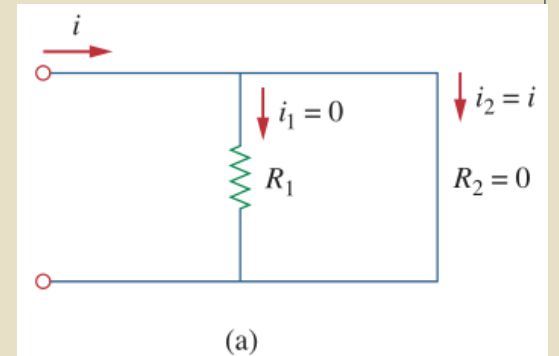
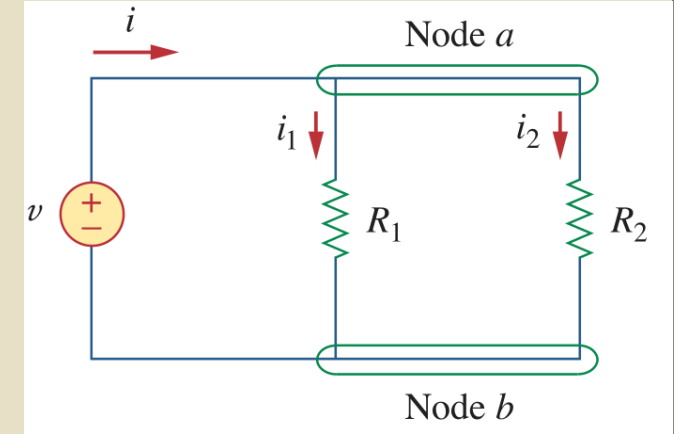
For the given circuit, $v = iR_{eq} = \frac{iR_1R_2}{R_1+R_2}$

Thus, $i_1 = \frac{R_2i}{R_1+R_2}$ and $i_2 = \frac{R_1i}{R_1+R_2}$

The current i is shared by the resistors in inverse proportion to their resistances. This is known as the principle of current division. And the circuit is known as current divider circuit.

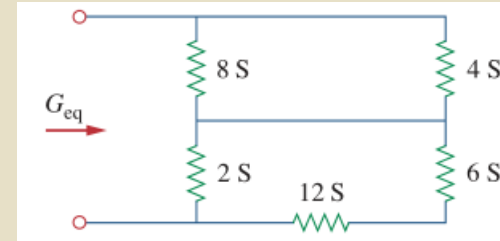
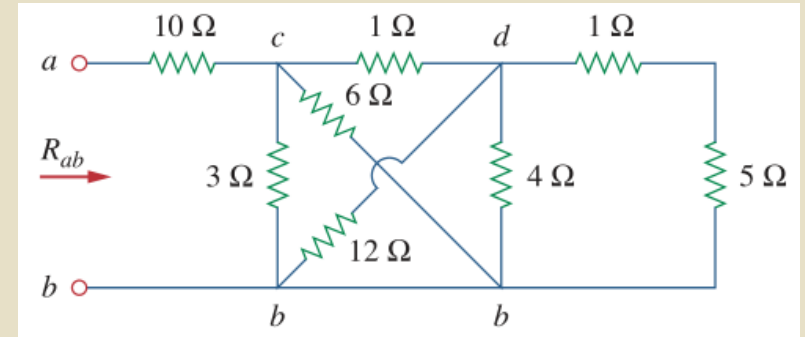
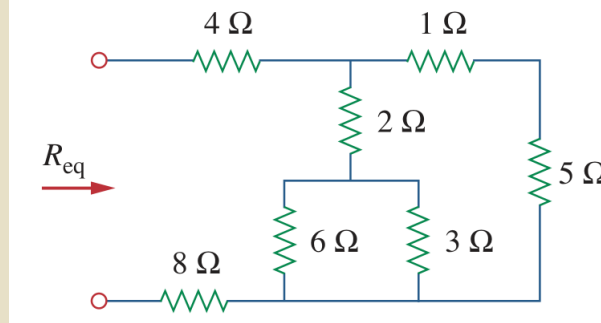
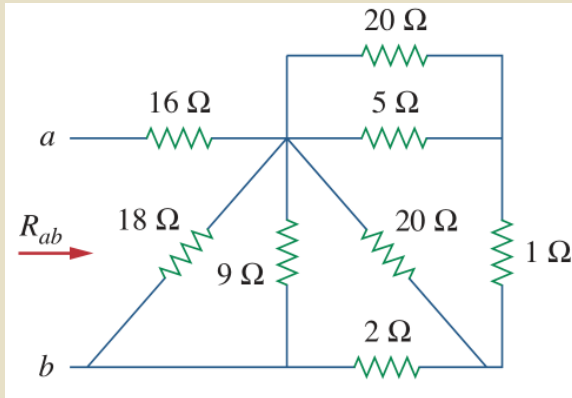
2nd circuit is a shorted circuit where we consider resistance is zero. Thus all of the current will bypass R_1 and flow through it. As current always flow to the least resistance path.

3rd circuit is opened circuit where we consider infinite resistance. Thus no current will flow through it.

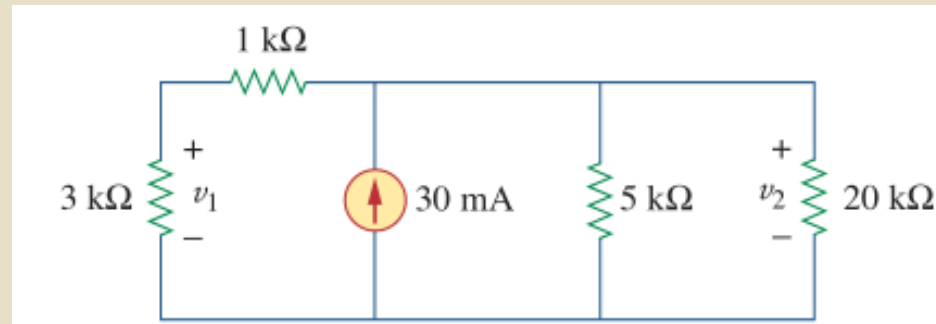
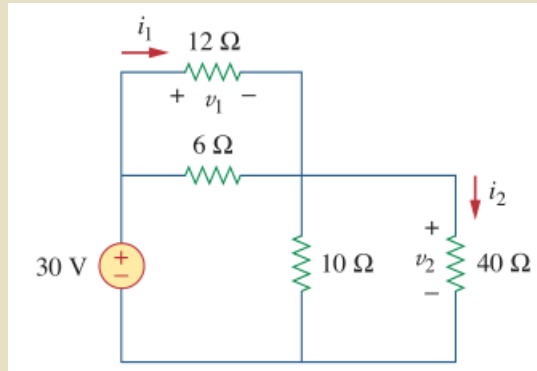


Practice Problems

- Find R_{eq} of the following circuits:



- Find V_1 and V_2 from the following circuits:



Wye Delta Transformation

- Situations often arise in circuit analysis when the resistors are neither in parallel nor in series. In this case Y-Delta Connection is used.
- Fig 1(a) and 1(b) shows wye(Y) or tee(T) network. And Fig 2(a) and 2(b) shows delta (Δ) or pi (Π) network.

They are used in three phase networks, electrical filters and matching Networks.

- $R_{12}(Y) = R_1 + R_3$

- $R_{12}(\Delta) = R_b || (R_a + R_c)$

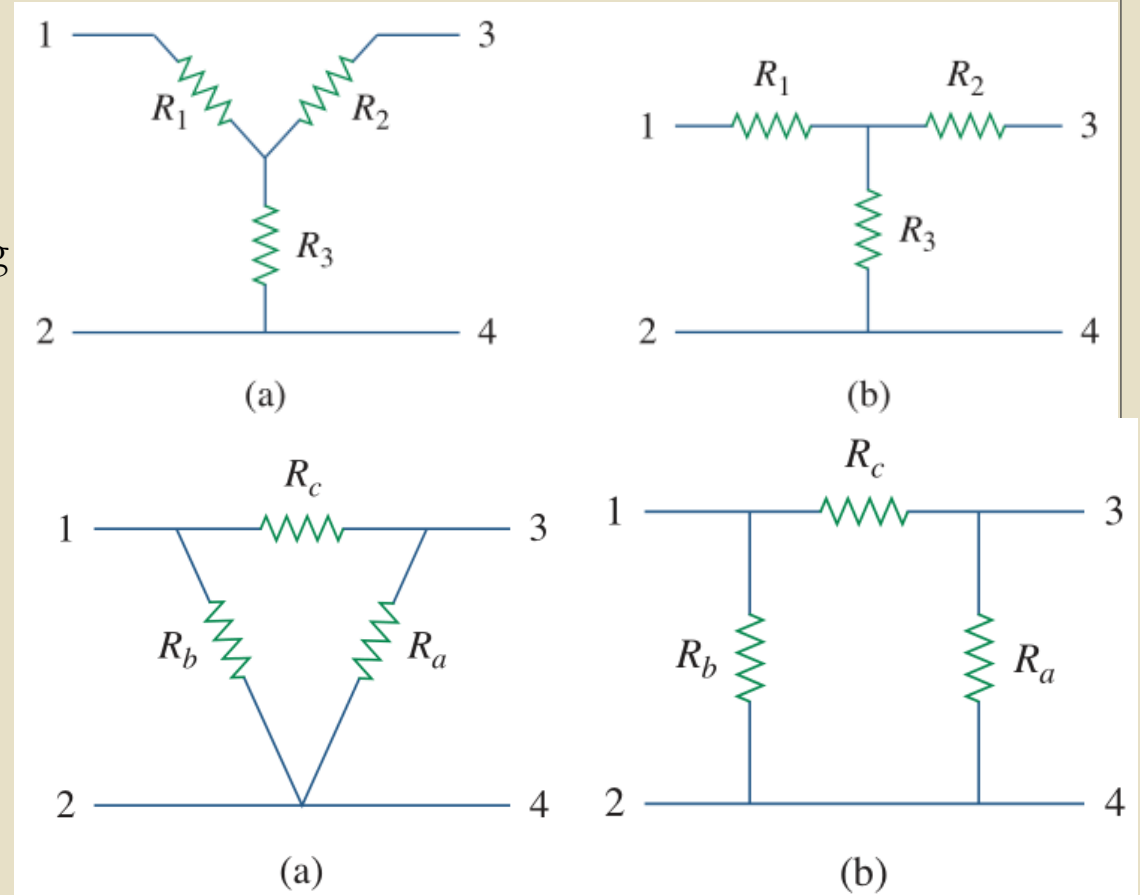
Setting $R_{12}(Y) = R_{12}(\Delta) = R_1 + R_3 = \frac{R_b(R_a + R_c)}{R_a + R_b + R_c}$

Similarly, $R_{13} = R_1 + R_2 = \frac{R_c(R_a + R_b)}{R_a + R_b + R_c}$

$$R_{34} = R_2 + R_3 = \frac{R_a(R_b + R_c)}{R_a + R_b + R_c}$$

Subtracting, $R_1 - R_2 = \frac{R_c(R_b - R_a)}{R_a + R_b + R_c}$

Thus, $R_1 = \frac{R_b R_c}{R_a + R_b + R_c}$; $R_2 = \frac{R_c R_a}{R_a + R_b + R_c}$; $R_3 = \frac{R_a R_b}{R_a + R_b + R_c}$



Delta to Wye Conversion

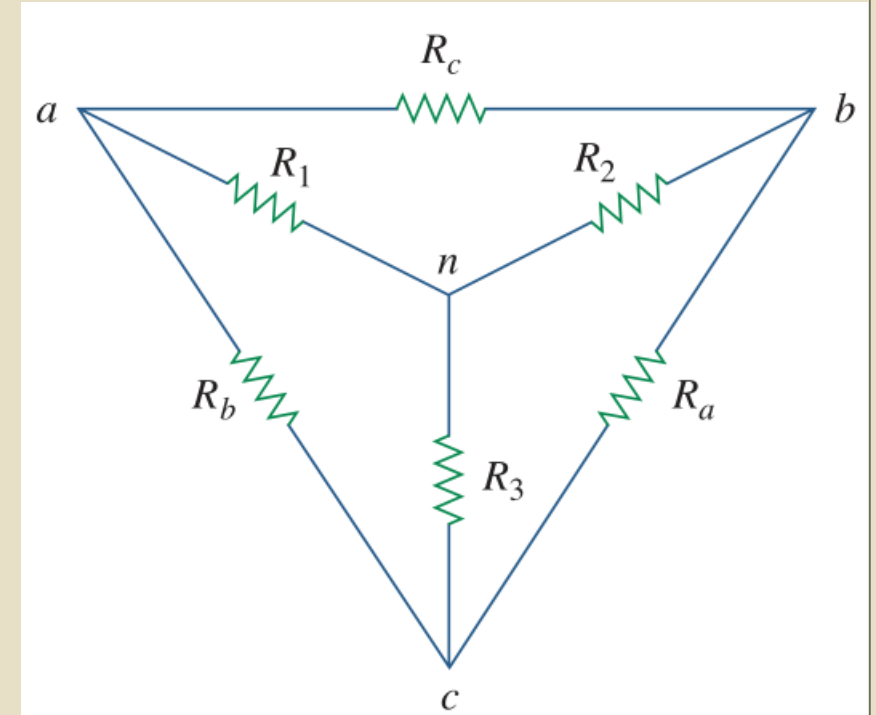
Each resistor in the Y network is the product of the resistors in the two adjacent Δ branches, divided by the sum of the three Δ resistors.

For the following circuit,

$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c}; R_2 = \frac{R_c R_a}{R_a + R_b + R_c}; R_3 = \frac{R_a R_b}{R_a + R_b + R_c}$$

If, $R_a = R_b = R_c = R_\Delta$

Then, $R_1 = R_2 = R_3 = R_Y = \frac{R_\Delta}{3}$



Wye to Delta Conversion

Each resistor in the Δ network is the sum of all possible products of Y resistors taken two at a time, divided by the opposite Y resistor.

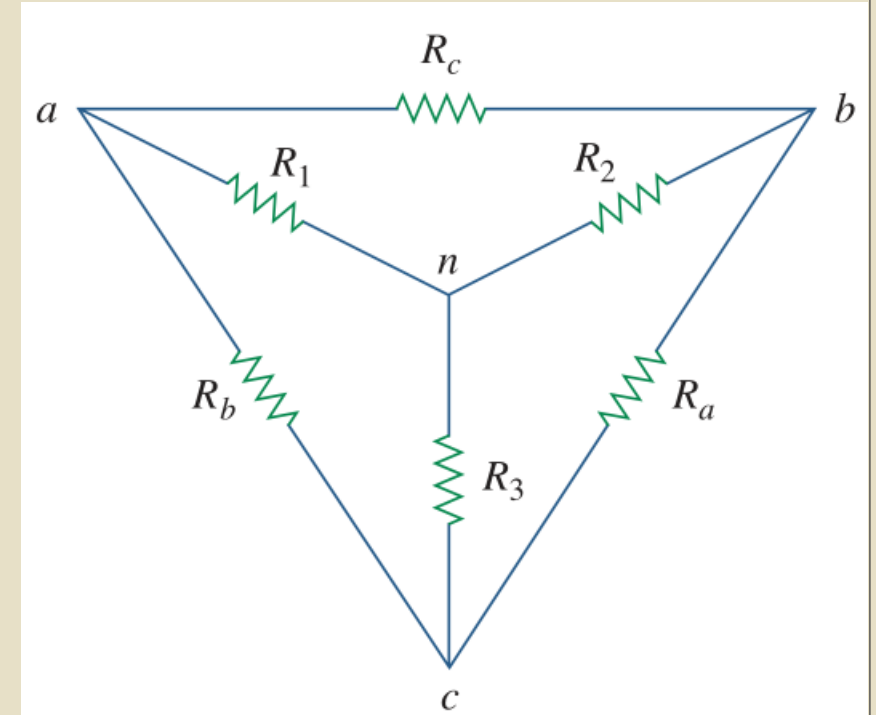
For the following circuit,

$$R_a = R_2 + R_3 + \frac{R_2 R_3}{R_1}; R_b = R_1 + R_3 + \frac{R_1 R_3}{R_2}; R_c = R_1 + R_2 + \frac{R_1 R_2}{R_3}$$

If, $R_a = R_b = R_c = R_{\Delta}$

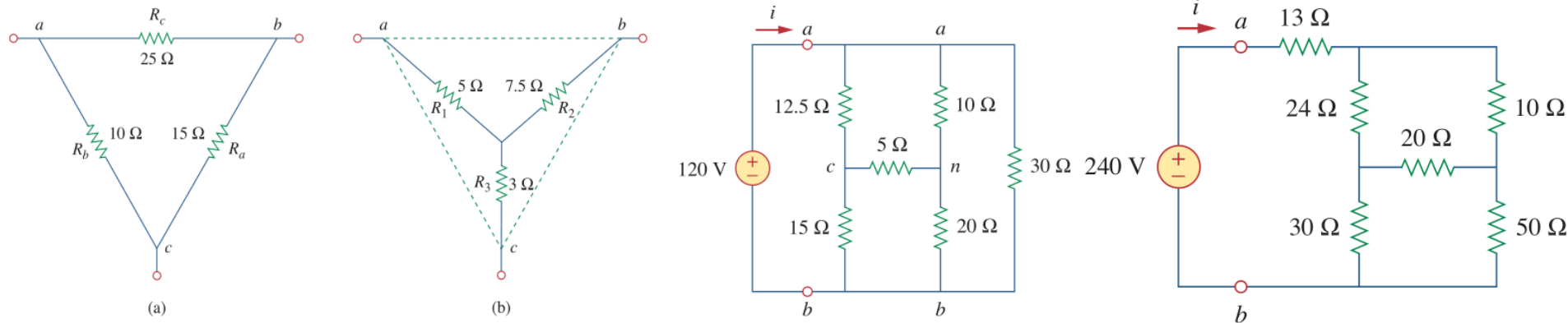
Then, $R_1 = R_2 = R_3 = R_Y = \frac{R_{\Delta}}{3}$

or, $R_{\Delta} = 3 R_Y$

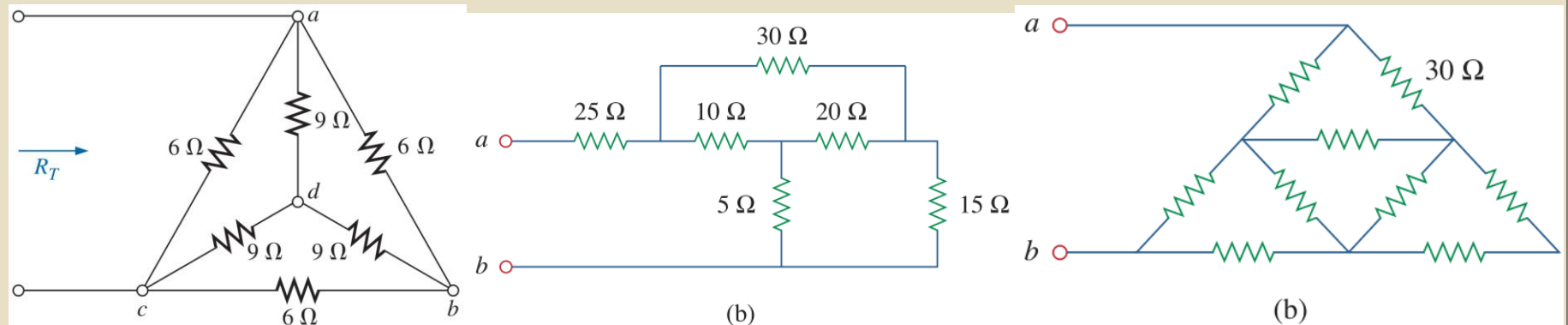


Wye to Delta Conversion

- Convert the given circuits into equivalent star/ delta form and find equivalent resistance.



- Find equivalent resistance using star delta conversion.



Next: Network Theorems